



Unit Test - Form A

Math and Measurement

STUDENT ASSESSMENT

Name: _____ Date: _____ Period: _____

Directions: Select the best answer for each question. Mark one choice only. Complete this test after all lessons and the cumulative review.

1. Which observation is quantitative?

- A. The solid is shiny.
- B. The sample mass is 12.40 g.
- C. The vapor has a sharp odor.
- D. The liquid appears clear.

2. Which tool is best for measuring 24.6 mL of water?

- A. Electronic balance
- B. Beaker
- C. Graduated cylinder
- D. Thermometer

3. Which record follows SI writing conventions?

- A. 0.45 g
- B. .45g
- C. 0.45 gs
- D. 0.45 G

4. What is 0.0825 L in milliliters?

- A. 0.0000825 mL
- B. 0.825 mL
- C. 8.25 mL
- D. 82.5 mL

5. Which is 0.0005070 in scientific notation?

- A. 5.07×10^{-3}
- B. 5.070×10^{-4}
- C. 50.70×10^{-5}
- D. 0.5070×10^{-3}



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6. How many significant figures are in 700.?

- A. 3
- B. 1
- C. 2
- D. 0

7. What is 12.4×3.20 with correct significant figures?

- A. 39.68
- B. 40
- C. 39.7
- D. 39.680

8. What is $18.42 + 3.7 + 0.006$ with correct precision?

- A. 22.126
- B. 22.13
- C. 22
- D. 22.1

9. Which factor converts 2.50 h directly toward seconds?

- A. 3600 s / 1 h
- B. 1 h / 3600 s
- C. 60 h / 1 min
- D. 1 s / 3600 h

10. What is 90.0 km/h in m/s?

- A. 2.50 m/s
- B. 9.00 m/s
- C. 25.0 m/s
- D. 32.4 m/s



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11. A sample has mass 57.0 g and volume 20.0 mL. What is its density?

- A. 0.351 g/mL
- B. 2.85 g/mL
- C. 37.0 g/mL
- D. 1140 g/mL

12. A liquid has density 1.24 g/mL and volume 25.5 mL. What is its mass?

- A. 0.0486 g
- B. 20.6 g
- C. 30.74 g
- D. 31.6 g

13. Experimental density is 12.4 g/mL; accepted density is 12.0 g/mL. What is percent error?

- A. 0.0333%
- B. 0.400%
- C. 3.33%
- D. 96.8%

14. Trials are 5.01, 5.00, and 5.02 g. Which claim is supported without an accepted value?

- A. The trials are precise.
- B. The trials are accurate.
- C. The method has zero error.
- D. The balance is calibrated.

15. A mean is very close to an accepted value. Which property is most directly supported?

- A. High resolution
- B. High precision
- C. Random error
- D. Accuracy



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16. What is the best repair for reading a concave meniscus from above?

- A. Use a wider container.
- B. Read at eye level at the bottom of the meniscus.
- C. Add more liquid.
- D. Round to fewer digits.

17. Why can percent error not establish precision by itself?

- A. Precision requires repeated trials.
- B. Percent error has no units.
- C. Accepted values are always exact.
- D. Precision uses only qualitative data.

18. A conversion ends with both km and m in the units. What should the student do?

- A. Round the number.
- B. Add the units.
- C. Ignore one unit.
- D. Reorient a factor so the unwanted unit cancels.

19. Samples of 5.0 g and 50.0 g of the same material have the same density. What does this show?

- A. Density is exact.
- B. Mass is intensive.
- C. Density is intensive.
- D. Volume is constant.

20. Which final record communicates a calculated density most completely?

- A. 2.85
- B. 2.85 g/mL
- C. about three
- D. density was measured